

# Resampling Techniques

# Problem 1: Empirical Bias Expansion

**Model:**  $X_1, \dots, X_n \sim \text{Exponential}(\lambda)$

**Statistic:**

$$\hat{\lambda} = \frac{1}{\bar{X}}$$

From theory:

$$\mathbb{E}[\hat{\lambda}] = \lambda + \frac{\lambda}{n} + O\left(\frac{1}{n^2}\right)$$

**Goal:** Empirically recover this bias expansion using simulation.

# Tasks (Problem 1)

- 1 Fix  $\lambda = 1$ .
- 2 For  $n \in \{5, 10, 20, 50, 100\}$ :
  - Simulate the sampling distribution of  $\hat{\lambda}$
  - Estimate empirical bias
- 3 Fit a regression model:

$$\widehat{\text{Bias}}(\hat{\lambda}_n) = \frac{c_1}{n} + \frac{c_2}{n^2}$$

- 4 Compare estimated  $c_1, c_2$  with theory.
- 5 Now take  $\lambda = \{1, 3, 5, 7\}$  and do the above 4 steps to check with theoretical values.

**Deliverable:** Bias vs  $1/n$  plot and fitted coefficients.

## Problem 2

Let  $X_1, \dots, X_n$  be i.i.d. with common distribution  $F$ .

We study the sampling distribution of the sample mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

The Central Limit Theorem (CLT) states that under suitable conditions:

$$Z_n = \frac{\bar{X} - \mathbb{E}[X]}{\sqrt{\text{Var}(X)/n}} \xrightarrow{d} \mathcal{N}(0, 1)$$

# Distributions Under Study

We compare four distributions:

- Normal(0, 1)
- Uniform(0, 1)
- Exponential(1)
- Cauchy(0, 1)

Each distribution exhibits different tail behavior and moment properties.

# Probability Densities (Reference)

**Normal(0, 1):**

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

**Uniform(0, 1):**

$$f(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

**Exponential(1):**

$$f(x) = e^{-x}, \quad x \geq 0$$

**Cauchy(0, 1):**

$$f(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R}$$

# Moments and CLT Applicability

| Distribution | Mean      | Variance  | CLT? |
|--------------|-----------|-----------|------|
| Normal       | finite    | finite    | Yes  |
| Uniform      | finite    | finite    | Yes  |
| Exponential  | finite    | finite    | Yes  |
| Cauchy       | undefined | undefined | No   |

**Key requirement for CLT:** Finite variance.

# Distance Measure I: Kolmogorov–Smirnov Distance

Let  $F_n$  be the empirical CDF of  $Z_n$ .

The KS distance from  $\mathcal{N}(0, 1)$  is:

$$d_{\text{KS}} = \sup_{x \in \mathbb{R}} |F_n(x) - \Phi(x)|$$

- Measures maximum vertical deviation
- Sensitive to global shape differences



## Distance Measure II: Wasserstein Distance

The Wasserstein-1 distance is defined as:

$$d_W = \int_0^1 |F_n^{-1}(u) - \Phi^{-1}(u)| du$$

- Measures average displacement between distributions
- Sensitive to tail behavior

## Tasks (Problem 2)

- 1 For each distribution and  $n \in \{10, 30, 100\}$ :
  - Simulate the sample mean  $\bar{X}$
  - Compute the standardized mean  $Z_n$
- 2 Plot empirical distributions of  $Z_n$
- 3 Compare with  $\mathcal{N}(0, 1)$
- 4 Quantify deviation using distance measures.
- 5 Comment on the convergence rates.